FREM: Causal Fourier–Reservoir Engram Memory for Financial Forecasting and Liability-Aware Allocation

Evidence-Bound Specification, Exploratory Findings, and the FREM-S14 Successor Protocol

Research specification prepared for Ryan Gomez

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Version 0.5: exploratory evidence and prospective FREM-S14 protocol

Primary memory advantage not established. Synthetic and public exploratory campaigns are complete. S3 memory effects are mixed and both primary intervals include zero. The prospective FREM-S14 GPU successor is implemented and packaged but remains numerically unqualified: its first GPU candidate failed a contraction test before market access, and its next saved version failed while Kaggle mounted the declared dataset before notebook code ran. Canonical admitted-market confirmation remains unevaluated; protected 2025 onward outcomes remain untouched. This draft reports actual evidence and does not establish profitability or submission readiness.

Abstract

Financial predictors must separate recurring information from transient noise while respecting information availability, execution timing, and investor liabilities. We specify FREM, a candidate architecture combining temporal joint-embedding prediction, contraction-bounded reservoirs, trailing Fourier descriptors, and bounded outcome-matured episodic memory. The proposed mechanism is delayed utility credit: an episode can influence retention and retrieval only after its outcome becomes observable. The FREM-S14 successor adds masked multi-offset EMA targets, an exact conditional finite-grid Gaussian diffusion transition, mature-neighbor graph retrieval, bounded training-residual-guided neurogenesis, and separately capped likelihood, downside, and tail routers. A constrained allocation layer consumes explicitly defined simple-return scenarios and protects investor cash-flow obligations. We formalize information timing, forward and training processes, frozen inference, uncertainty, cost accounting, and a liability case based on Laura Gao's supplied preparation document. Algebraic results establish reservoir contraction, causal forecast measurability under stated assumptions, and bounded retained state; they do not establish market predictability. The fixed S14 development family contains 19 arms, three seeds, and three expanding historical-proxy folds, with complete-family inference and an untouched 2025 onward boundary. The retained synthetic integration exercise favors the memory-off control. Completed public exploratory S2, six-window, S3, and S4 campaigns provide mixed or unsupported memory evidence; S3 primary gains are -0.014116 and +0.007006 nats in 2023 and 2024, with both intervals including zero. S14 has produced no market fit: one GPU engineering attempt failed a strict numerical test, and a later attempt failed during external dataset mounting before notebook execution. Canonical confirmation therefore remains unevaluated, and completed protocols remain closed.

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1 Research question and scope

The proposed scientific question is narrow: *after controlling information, preprocessing, forecast horizon, and declared training resources, does delayed utility-weighted spectral memory improve out-of-time predictive log score over the same JEPA–reservoir predictor without memory?* A secondary question asks whether any predictive gain survives execution delays, costs, concentration limits, and liability requirements. These are separate claims: improved likelihood does not imply improved trading returns, and improved historical returns do not establish future funding reliability.

The motivating mechanism is that a stable latent encoder can summarize common dynamics while an episodic store preserves unusual but recurring local contexts. Fourier descriptors characterize a trailing window's distribution across frequencies. A reservoir supplies a fixed nonlinear summary of its ordered history. The episodic store retains outcomes and prediction residuals associated with those descriptors. Merely being unusual is insufficient: delayed predictive utility also affects retention. This suggests a discriminating comparison against an equally bounded first-in-first-out store and uniform eviction.

The proposed practical advantage is supported only if the confidence interval clears the prespecified likelihood improvement threshold. An interval at or below zero supports a negative conclusion for the direction of improvement; an interval spanning the practical threshold can remain inconclusive. An advantage that disappears against simpler retention rules weakens the proposed retention mechanism. A methodological defect invalidates the test rather than falsifying the scientific hypothesis. A successful return forecast with noninformative memory would not support the proposed memory mechanism. A sufficiently precise negative result would still determine that this version's extra mechanism is unnecessary in the tested regimes.

The package provides executable research components rather than an assurance that arbitrary data can be made meaningful. Its main forecasting interface requires finite, positive total wealth observations, causal availability metadata, and an explicit executable-observation convention. It supports mapped tabular time-series panels and optional vintage macro features. Images, raw documents, broker order books, missing legal permissions, and zero-recovery defaults are not silently converted into equivalent inputs. A default or delisting with zero terminal recovery needs a separate point-mass outcome model; excluding it to make a backtest pass would invalidate the study.

2 Prior art and the novelty boundary

I-JEPA learns representations by predicting target embeddings from context embeddings; its original setting is visual representation learning [1]. Temporal JEPA architectures already exist, including TS-JEPA [7]. More recent work reports that the benefit of a particular JEPA instantiation can differ substantially across forecasting backbones [9]. These sources motivate a JEPA-off control and prevent treating the auxiliary objective as inherently beneficial.

Echo-state networks use fixed recurrent dynamics and trained readouts, with stability conditions that must be distinguished from empirical reservoir heuristics [13]. Combining reservoir methods and Fourier representations is also established territory: multi-scale random Fourier reservoirs and frequency-domain reservoir computing are close conceptual neighbors [15, 18]. FREM's dense time-domain reservoir plus a trailing DFT is not the frequency-domain recurrence in FRESCO, and no efficiency result from that work transfers automatically.

Engram introduces n-gram-based conditional lookup for language models [5]. FREM uses the more modest concept of quantized short temporal motifs and an explicit episodic store; it does

not reproduce Engram's large static embedding architecture or its constant-time lookup claim. FREM's exact memory search scans its bounded record set and then sorts candidates. Its retrieval cost is therefore explicit rather than assumed negligible.

VICReg motivates variance and covariance penalties against representation collapse [3]. FREM uses related penalties on context embeddings together with an EMA target, and must measure collapse directly. The feedback controller is a bounded learning-rate, auxiliary-weight, and expert-weight policy. It does not learn a new differentiation operator. Continual backpropagation, which replaces underused units, is a distinct existing idea [6]; it is not claimed as implemented by the present controller.

XGBoost is gradient tree boosting [4]. It serves as a standalone benchmark and an optional residual corrector. It neither traverses a financial graph nor makes the reservoir run faster. DLinear motivates a simple decomposition-based linear sequence control, while patch-based Transformers motivate a compact tokenized sequence control [16, 24]. The package's task-specific DLinear and patch-Transformer implementations are architectural controls, not claimed reproductions of published state-of-the-art scores.

Financial retrieval and memory are already explicit research directions. FinMem combines layered memory with an LLM trading agent [23]. FinSrag retrieves financial time-series segments using FinSeer and supplies them to StockLLM [22]. These are direct application-level neighbors, although their language-model interfaces differ from FREM's numerical residual memory. Their existence rules out a broad claim to originate financial memory or retrieval. FREM's narrower question concerns outcome-matured utility retention under matched information and bounded storage. That distinction is a candidate mechanism, not established novelty, and neither publication's reported performance transfers to FREM.

Repeatedly choosing a strategy because it looks good on the same historical outcomes can overfit the backtest [2]. Consequently, the package separates development from protected confirmation, retains failed runs, and does not treat a new seed as a new market history. The literature review defines important neighbors but does not establish exhaustive novelty. Before submission, full-text mechanism comparisons and matched-information experiments against suitable retrieval controls remain necessary; the added primary-source records do not constitute an exhaustive novelty audit.

Component	Established idea	Candidate contribution to test
JEPA	Predict future latent representations	Whether that objective helps this specific predictor
Reservoir and Fourier	Fixed recurrence and frequency features	Whether their stable descriptors improve episode selection
Engram memory	Local-pattern lookup and episodic retrieval	Delayed predictive utility governing bounded retention
Adaptive controller	Feedback and expert weighting	Whether the declared bounded updates improve stability
Stochastic allocation	Scenario risk and constrained optimization	Whether forecasts help after all funding and trading constraints

Table 1: The research contribution is not the mere coexistence of seven familiar components.

3 Information, observations, and targets

Let $i \in \{1, \ldots, A\}$ denote an instrument and t a UTC decision timestamp. The available information $\mathcal{F}_t$ contains only observations published by t, the instrument master and eligibility known at t, and previous matured outcomes. Event time and availability time are separate:

a monthly observation stamped January is not available at January's start merely because it describes that month. Revised macro observations require their vintage publication times.

The input $X_{i,t} \in \mathbb{R}^{L\times D}$ is the trailing observed context, accompanied by an ordered feature schema and missingness channels. The future training target context $X^{+}_{i,t} \in \mathbb{R}^{H\times D}$ is accessed only inside fitting. The public inference call receives $X_{i,t}$ and never uses $X^{+}_{i,t}$ or its associated future return to construct the current forecast.

The implemented default is next-executable-close research accounting. Let $e_{i,t}$ be the first tradable observation with event time strictly after the decision and let $q_{i,t}$ be the observation H executable intervals later. This conservatively prevents using the close that created the signal as a free simultaneous entry. It does not guarantee an actual fill at the next close. A different entry convention requires a separately specified dataset and protocol version.

With V_i the appropriately reconstructed total-wealth series in base currency,

$$Y_{i,t} = \log \frac{V_i(q_{i,t})}{V_i(e_{i,t})}, \qquad u_{i,t} \geq q_{i,t}, \tag{1}$$

where $u_{i,t}$ is the outcome-availability timestamp, including the relevant price, distribution, corporate-action, and FX availability. Market returns, forecast locations, and predictive scales use this same horizon. The separate allocation layer requires *simple* returns, not these log-return locations.

The currency identity is

$$1 + R^{\text{base}}_{i,t} = (1 + R^{\text{local}}_{i,t})(1 + R^{\text{FX}}_{i,t}). \tag{2}$$

The FX convention is base-currency units per local-currency unit. FX observations must themselves be available at the decision or realization being evaluated. Bond yields are not bond prices or total returns: a yield series alone is rejected as the target asset-wealth series. Coupons, accrued interest, maturity, defaults, and distribution reinvestment require consistent provider-specific treatment.

3.1 Point-in-time preprocessing

Schema mapping renames explicitly supplied columns; it does not infer economic semantics from arbitrary numbers. Validation rejects duplicate instrument/event rows, inconsistent ordering, unavailable eligibility, unsuitable prices, missing corporate-action provenance, and unsupported FX assumptions. Custom covariates need a documented availability convention. The original source, license, retrieval date, transformations, and hashes belong in the data manifest.

For feature d, the training-context median m_d and interquartile range s_d define

$$\widetilde{X}_{\tau d} = \operatorname{clip}\left(\frac{\operatorname{impute}(X_{\tau d}; m_d) - m_d}{s_d}, -12, 12\right). \tag{3}$$

Constant features use unit scale; all-missing training features use zero median and are recorded. Binary missingness channels are preserved without continuous rescaling. Statistics are fitted once per training fold, including the repeated observations that appear in overlapping training contexts, and then frozen. No backward fill uses later prices to repair an earlier observation. The package also retains the unscaled contexts for reproducible reconstruction.

Optional peer features aggregate other instruments' latest available return snapshots, conditioned on market and sector classification known at the decision. They exclude the focal instrument, enforce a staleness limit, and require a minimum peer count. These features create cross-asset context. They are not a learned financial graph, and pairwise full-sample correlations are not computed as historical features.

4 Architecture and complete forward specification

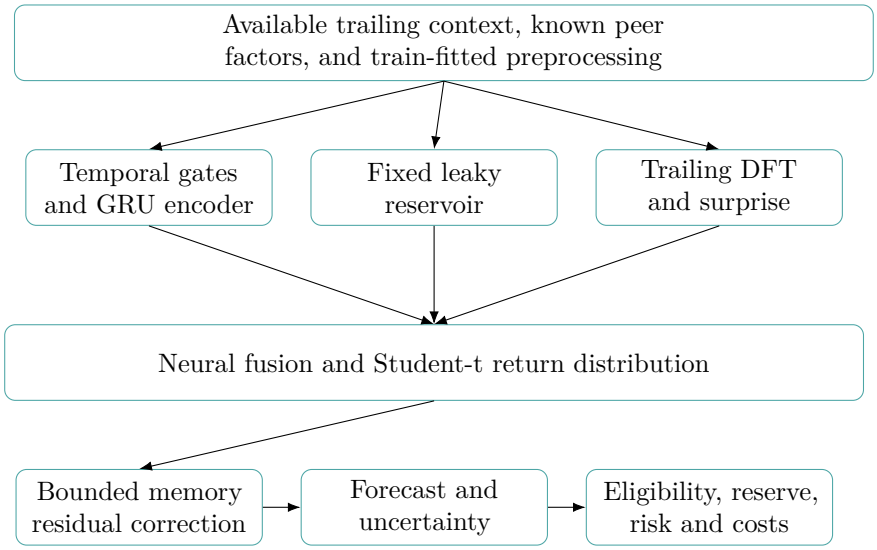


Figure 1: Forecast-time paths. Future contexts appear only in the training JEPA loss. Memory utility changes only after outcomes mature. Optional XGBoost residual correction is calibrated on a separate chronological interval.

4.1 Temporal representation and relevance gates

For each observed prefix, let $\bar{x}_\tau = \tau^{-1} \sum_{j \le \tau} \widetilde{x}_j$. The feature gate and the temporal gate are

$$g_\tau = 2\sigma(A[\widetilde{x}_\tau; \bar{x}_\tau] + b), \qquad a_\tau = 2\sigma\{\mathrm{Conv}_3^{\mathrm{left-pad}}(\widetilde{X})_\tau\}. \tag{4}$$

A unidirectional GRU consumes $g_\tau \odot \widetilde{x}_\tau a_\tau$. An attention-weighted average of its hidden sequence is concatenated with its final hidden state, projected through a linear layer, GELU, dropout, and layer normalization, yielding $z_t \in \mathbb{R}^d$. All recurrent and convolutional dependencies lie inside the available trailing context. Attention pooling over the complete trailing window is causal at decision time even though it combines all its historical positions.

The gate-off control sets the input modulation to one and omits the learned gates. Channel relevance does not prove causal economic importance. Correlated peer features can substitute for each other, and learned gate magnitudes are diagnostic quantities rather than treatment effects.

4.2 Contraction-bounded reservoir

The fixed reservoir receives the scaled raw context, independently of the learned temporal gate:

$$r_\tau = (1-\alpha) r_{\tau-1} + \alpha \tanh(W r_{\tau-1} + U \widetilde{x}_\tau + b_r), \quad r_0 = 0. \tag{5}$$

The recurrent random matrix is rescaled so $\|W\|_2 = \rho < 1$, with defaults $\alpha = 0.3$, $\rho = 0.9$, and reservoir dimension $R = 64$. Input weights have variance normalized by input dimension. Recurrent weights and biases are fixed buffers. Only the downstream fusion learns how to use r_L. The reservoir is reset for each context; it does not secretly accumulate an unlimited asset history.

Proposition 1 (Forgetting of initial state)**.** *For identical inputs and any two initial states, the reservoir obeys*

$$\|r_\tau - r'_\tau\|_2 \le (1 - \alpha + \alpha\rho)^\tau \|r_0 - r'_0\|_2.$$

Proof. The componentwise hyperbolic tangent is 1-Lipschitz in Euclidean norm. Apply the triangle inequality to the recurrence and $\|Wv\|_2 \le \rho\|v\|_2$, then iterate. With the defaults the one-step bound is 0.97. This is an algebraic bound, not a measured performance statistic. □

The result justifies bounded sensitivity to initialization under these assumptions. It does not imply financial sample efficiency, identification of hidden market states, or recoverability of information absent from the input. A strong contraction may itself erase useful long-range structure.

4.3 Fourier descriptors and spectral surprise

For $j = 0, \dots, L-1$, apply the Hann window $h_j = (1 - \cos(2\pi j/(L-1)))/2$ to each demeaned trailing feature. The ordinary unitary real-input DFT is

$$C_{kd} = L^{-1/2} \sum_{j=0}^{L-1} h_j(\widetilde{X}_{jd} - \bar{X}_d) e^{-2\pi \mathrm{i} kj/L}, \quad 0 \le k \le \lfloor L/2 \rfloor. \tag{6}$$

Partition these bins into B contiguous, approximately equal-sized bands. For each band b and feature d, store the band averages of $\log(1 + \mathrm{mean}_{k\in b}\,|C_{kd}|^2)$, real amplitude, and imaginary amplitude. More precisely, the logarithm is taken after band averaging power. Training-context means μ^P_{bd} and variances v^P_{bd} of the log-power define

$$S_{bd} = \min\left(20, \frac{|P_{bd} - \mu^P_{bd}|}{\max(\sqrt{v^P_{bd}}, 10^{-3})}\right). \tag{7}$$

The variance floor is 10^{-6}. Learnable band weights $w_b = 2\sigma(\beta_b)$ multiply the three raw descriptor channels and the surprise channel. Concatenation yields $4BD$ neural features. Mean surprise is also recorded as a candidate episode's novelty score.

This is a modified *use* of a Fourier representation: a standard DFT followed by adaptive band weighting and a separate memory rule. No new transform or inversion theorem is claimed. Demeaning, a trailing window, and phase-sensitive channels do not make irregular calendar sampling equivalent to uniform physical time. The current DFT uses ordered observation indices; for markets with unequal holiday gaps, a frequency means cycles per observation. A continuous-time spectral interpretation would require a different explicit method.

4.4 N-gram and fixed memory keys

The most recent $p \le 8$ observations of a selected scaled feature are quantized using thresholds $(-1, -0.25, 0.25, 1)$ into five symbols. These fixed thresholds avoid fitting a codebook on future observations. The symbol sequence is encoded as a base-5 integer and as concatenated one-hot vectors. The default is $p = 3$ with the log-return feature selected by its documented index.

A memory key concatenates separately ℓ_2-normalized blocks: the latest scaled feature vector, its trailing mean, the final fixed reservoir state, the *unweighted* raw spectral descriptor, and the n-gram one-hot representation. The concatenation is normalized again and detached from gradient computation. Turning off a branch removes its corresponding key block. Key dimension is

$$D_k = 2D + \mathbf{1}_{\text{reservoir}} R + \mathbf{1}_{\text{spectral}} 3BD + \mathbf{1}_{\text{ngram}} 5p. \tag{8}$$

Frozen preprocessing, fixed reservoir buffers, and unweighted descriptors prevent learned-encoder drift from silently changing key meanings. Stored prediction residuals can still age as the predictor changes during training. Fixed keys solve one drift problem, not every nonstationarity problem.

4.5 Retrieval, correction, and retention

An episode stores identifier, key k_j, instrument, decision time t_j, availability u_j, realized log return y_j, residual $e_j = y_j - \mu_j^0$, n-gram code, surprise s_j, utility U_j, and credited reuse count c_j. A record is eligible at t only when $u_j < t$ and $t_j < t$. Same-instrument retrieval is the default; exact n-gram matching is optional. Cross-instrument retrieval must be separately declared.

Among eligible records, choose the K largest cosine similarities. Stable identifier order breaks ties. The retrieval weight is

$$\omega_j = \mathrm{softmax}_{j \in \mathcal{N}_t} \left(\frac{k_t^\top k_j}{T_m} - \log(2) \frac{t - u_j}{h_m} + \tanh(U_j) \right), \tag{9}$$

where ages and half-life h_m share units of days, with defaults $T_m = 0.2$, $h_m = 180$, $K = 8$. All retention controls use the same similarity, age, and utility retrieval weights; only eviction changes. Memory supplies seven summaries: weighted residual mean, outcome mean, residual standard deviation, cosine similarity, $\log(1+K_t)$, utility, and a bounded age statistic. An empty store produces seven zeros.

The neural fusion concatenates z_t, the available reservoir state, and spectral features, then applies a linear layer, GELU, dropout, and layer normalization. Let this representation be f_t. The final memory gate is

$$\gamma_t = \mathbf{1}_{K_t > 0}\, q_t\, \sigma\{G([f_t; m_t])\}, \qquad \mu_t = \mu_t^0 + \gamma_t \sum_{j \in \mathcal{N}_t} \omega_j e_j, \tag{10}$$

where $q_t \in [0, 1]$ is a controller weight. Disabling memory yields $\gamma_t = 0$. The memory gate's parameters remain present in that control, although they are inactive; actual compute differences are reported.

At a decision, the predictor schedules a pending event with its key, recorded forecasts, uncertainty, retrieved identifiers, and weights. At a later time strictly after the outcome becomes available, it computes

$$\Delta_t = \mathrm{clip}\left(\frac{(y_t - \mu_t^0)^2 - (y_t - \mu_t)^2}{s_t^2 + (y_t - \mu_t^0)^2 + 10^{-12}}, -1, 1 \right). \tag{11}$$

Retrieved episode j receives $U_j \leftarrow (1 - \eta_u \omega_j) U_j + \eta_u \omega_j \Delta_t$ with $\eta_u = 0.1$, and its credited count increases. This is shared credit for the used correction, not a counterfactual causal attribution to an individual memory.

The new matured episode is inserted if its surprise meets the configured write threshold, which defaults to zero. Thus the default permits ordinary observations as well as anomalies. When the capacity $C = 512$ is exceeded, utility retention removes the record with the smallest

$$Q_j = \tanh(U_j) + 0.1 \tanh(s_j) + 0.03 \log(1 + c_j) - \log(2) \frac{t - u_j}{h_m}. \tag{12}$$

FIFO and uniform-eviction controls preserve the declared capacity. Uniform eviction among current candidates is not uniform reservoir sampling over the entire history. Pending events have a separate explicit capacity, default $P = 100{,}000$; exhaustion raises an error rather than silently discarding labels.

4.6 Distributional prediction and stochastic interpretation

Three return-head outputs a_t, b_t, c_t define

$$\mu_t^0 = Ha_t, \qquad s_t = \sqrt{H}\,\mathrm{softplus}(b_t) + s_{\min}, \qquad \nu_t = 2.1 + \mathrm{softplus}(c_t), \tag{13}$$

with $s_{\min} = 10^{-4}$. The predictive density is $Y_t \mid \mathcal{F}_t \sim t_{\nu_t}(\mu_t, s_t)$ and $\mathrm{Var}(Y_t \mid \mathcal{F}_t) = s_t^2\nu_t/(\nu_t - 2)$. Scale is not standard deviation. The default head starts with zero return location, a 1% step scale, and approximately five degrees of freedom. These are initialization choices, not estimated market conditions.

The H and $\sqrt{H}$ factors give a convenient drift/scale parameterization, but they do not prove temporal aggregation consistency for dependent returns or a Student-t diffusion. The fixed-scale control trains squared error, uses a calibrated constant scale, and a high-degree Student-t approximation to a Gaussian for scoring.

Proposition 2 (Log-Student-t arithmetic-mean limitation)**.** *For a nondegenerate Student-t random variable Y with finite degrees of freedom, $\mathbb{E}[e^Y] = \infty$.*

Proof. Its positive-tail density decays polynomially, proportional to $y^{-(\nu+1)}$. For sufficiently large y, the integral of e^y times any such polynomial tail diverges. Translation and positive scaling do not change this conclusion. □

Consequently, $\mathrm{expm1}(\mu_t)$ is a transformed location, not an expected simple return. The allocation interface refuses an undocumented substitution. One may use explicitly bounded return scenarios or a separately declared Gaussian-log-return model with finite exponential moments; either changes the distributional assumptions and must be recorded.

5 Training and adaptive learning

An EMA copy $\bar{\theta}$ of the temporal encoder defines the training-only target $z_t^+ = E_{\bar{\theta}}(X_t^+)$. The online latent predictor P_θ is a two-layer MLP. With unit-normalized embeddings,

$$\mathcal{L}_{\mathrm{JEPA}} = \frac{1}{nd}\sum_t \left\|\mathrm{norm}(P_\theta(z_t)) - \mathrm{stopgrad}\{\mathrm{norm}(z_t^+)\}\right\|_2^2, \tag{14}$$

where d is latent dimension and the implemented mean-squared error averages both observations and latent coordinates. Context-embedding variance is encouraged toward at least 0.5 standard deviation per dimension, with a 10^{-4} stabilizer. The covariance penalty is the sum of squared off-diagonal sample covariances divided by latent dimension. For singleton batches these terms are unidentifiable and explicitly omitted; the trainer accumulates multiple chronological groups when possible.

The supervised loss is the negative log density

$$\ell_t = \log s_t + \tfrac{1}{2}\log(\nu_t\pi) + \log\Gamma(\nu_t/2) - \log\Gamma((\nu_t+1)/2) \tag{15}$$

$$+ \tfrac{\nu_t+1}{2}\log\left(1 + \frac{(y_t-\mu_t)^2}{\nu_t s_t^2}\right). \tag{16}$$

The default objective is $\bar{\ell} + 0.1\mathcal{L}_{\mathrm{JEPA}} + 0.01\mathcal{L}_{\mathrm{var}} + 0.001\mathcal{L}_{\mathrm{cov}}$. AdamW uses learning rate 10^{-3}, weight decay 10^{-4}, and gradient-norm clipping at one. The target update after a neural optimizer step is $\bar{\theta} \leftarrow 0.99\bar{\theta} + 0.01\theta$. Target parameters receive no direct gradient, and target dropout is disabled.

Each epoch resets the episodic state and processes training decision times in order. Before a time group is forecast, previously scheduled outcomes that matured strictly earlier are released. All rows at the same decision time retrieve from the same pre-group store. Neural updates use only the training partition; validation affects the explicitly declared controller and checkpoint selection. Multiple training epochs can of course revisit the training labels, so their in-sample forecasts must not be relabeled as genuine historical out-of-sample predictions.

The learning controller has two restricted functions. First, after a prespecified number of unimproved validation scores, it reduces learning rate by a factor of 0.5, down to 10^{-6}, and reduces the JEPA coefficient down to a quarter of its original value. Second, matured comparisons between the base forecast and the memory-expert forecast update a bounded expert logit. The logit is clipped to $[-4, 4]$ and mapped through a sigmoid. It determines q_t in the memory correction. These mechanisms select the strength of predefined learning components; they do not invent architectures or optimize an arbitrary new learning algorithm.

The best validation checkpoint restores neural weights, optimizer, controller, and memory state. Validation uses the same asset-then-date balancing as the primary score; the minibatch training loss remains an observation mean. Any validation labels used to choose that checkpoint count as fitting information. Public forecasts must occur strictly after their recorded fitting-information cutoff. This rule also covers calibration labels used by the optional residual tree. A run that exhausts its budget before completing a requested validation cannot be reported as a valid selected checkpoint.

5.1 XGBoost roles and fairness

The standalone tree baseline consumes training-only tabular summaries of the same contexts. The optional hybrid fits XGBoost to residuals on a separate later calibration segment, using frozen neural predictions and context summaries. For residual estimate b_t, its correction is exactly $0.25\,\mathrm{clip}(b_t, -0.1, 0.1)$, with maximum absolute cumulative log-return correction 0.025. These fixed choices are development candidates, not optimized numerical claims.

The hybrid's tuning and calibration intervals are chronologically separated and purged within the original validation interval. It changes the use of those labels and performs an extra supervised residual fit; it does not necessarily receive more distinct observations than the basic neural candidate. It therefore belongs in an explicitly identified auxiliary comparison. The strongest information-matched claim concerns FREM ablations: classical and neural baselines use a purged internal training tail for scale calibration, whereas FREM uses the external validation interval for controller updates and checkpoint selection. Identical per-cell time caps do not by themselves equalize achieved optimization or selection opportunities. An apparent win is not evidence for memory unless these differences have been assessed. No silent substitute is used when XGBoost is unavailable: the dependency error is retained.

6 Inference, online feedback, and mathematical guarantees

Frozen inference loads the model, training scaler, spectral reference, fixed reservoir, memory, controller, ordered feature names, and asset mapping. It checks that decisions follow all fitting information, builds only causal context windows, retrieves eligible memory, and outputs location, scale, degrees of freedom, diagnostics, and timestamps. Unlabeled forecasts use unavailable-target markers; those markers cannot be interpreted as observed zero returns.

An explicit adaptive-inference option may release prior matured outcomes and update the episodic store and expert weight. It does not silently backpropagate through the protected evaluation set. Actual neural retraining would require a separately scheduled walk-forward protocol.

A new feedback observation must identify its original decision, instrument, realized outcome, and verified availability. Conflicting corrections require an audited replay, not overwriting historical results.

Proposition 3 (Causal forecast under a fitting cutoff)**.** *Assume the fitted model, scaler, reference, initial memory, and controller depend only on information available by T_0. For every decision $t > T_0$, assume context and metadata are $\mathcal{F}_t$-measurable, and memory updates use only outcomes with $u < t$. Then the forecast produced at t is $\mathcal{F}_t$-measurable.*

Proof. The initialization is measurable by assumption. Induct over ordered decision-time groups. The update at a group reads only already available outcomes. Eligibility, deterministic key formation, gated neural maps, and retrieval are functions of the current context and that updated state. They therefore preserve measurability. No same-group future outcome is released. The resulting state satisfies the induction hypothesis for the next group. □

This guarantee is conditional on truthful provider timestamps and on not training a checkpoint using later outcomes. A false availability column defeats the assumptions even if the software's comparisons are implemented correctly.

Proposition 4 (Bounded retained state)**.** *With episode capacity C, pending capacity P, and fixed key dimension D_k, episodic and pending storage are $O((C + P)D_k)$, apart from fixed-size metadata and model parameters.*

Proof. Insertion evicts whenever retained episodes exceed C. Scheduling rejects new events when the pending queue reaches P. Each record contains a fixed-length key and bounded-size retrieval metadata for fixed K. Thus the number and size of records have the stated bound. □

Stored training histories and experiment artifacts may grow with declared epochs or runs and are separate from online memory. Bounded memory does not imply bounded information loss or constant-time retrieval.

7 Risk scenarios, markets, and constrained allocation

The portfolio layer selects among the supplied eligible instruments. Their market, sector, asset class, currency, settlement, lot, liquidity, eligibility, and rule provenance form the allocation context. Market and sector selection occur through constrained instrument weights and their group totals. An absent market does not become investable merely because a forecast looks attractive.

7.1 From saved marginal forecasts to portfolio inputs

The executable forecast-to-portfolio bridge requires one common forecast time, stable ordered instrument identifiers, saved (μ_i, s_i, ν_i), a declared base currency, and an attested common future entry and exit interval. All input provenance must be dated no later than the decision. An identical count of trading bars across different calendars is insufficient to establish a common economic horizon. The operator must verify that synchronization before the bridge is used.

Let C be an explicitly supplied, dated latent Gaussian correlation matrix, or the identity only when independence is explicitly declared. For each scenario j,

$$Z_j \sim \mathcal{N}(0, C), \quad U_{ji} = \Phi(Z_{ji}), \quad R_{ji} = \operatorname{expm1}\left[\operatorname{clip}\{\mu_i + s_i T_{\nu_i}^{-1}(U_{ji}),\, a_i, b_i\}\right], \tag{17}$$

where a_i, b_i are the prespecified finite log-return bounds. Numerical CDF endpoints are clamped to machine epsilon and one minus epsilon before quantile evaluation; those events are counted. The clipping operation creates probability masses at the bounds: it is winsorization, not a conditional truncated Student-t distribution. Gaussian latent correlation is neither simple-return correlation nor a Student-t copula, and its joint tail behavior is not learned by the marginal predictor.

With nonnegative declared scenario probabilities p_j summing to one, the portfolio receives the finite-distribution moments

$$\widehat{m} = \sum_j p_j R_j, \qquad \widehat{\Sigma} = \sum_j p_j (R_j - \widehat{m})(R_j - \widehat{m})^\top. \tag{18}$$

There is no annualization or Bessel correction: these are exactly the moments of the retained discrete scenario distribution. Uniform Monte Carlo weights are the default; alternative weights need an explicit rationale. The decision artifact retains the scenarios, probabilities, forecast records, units, common horizon, bounds, clipping counts, dependence source, and content hashes. The bridge is one synchronous research allocation, not an executed trade or a complete portfolio backtest.

The separate raw-moment allocation interface requires an explicit simple-return, base-currency and common-horizon contract. Missing forecasts remain missing rather than becoming nonfinite array entries, allowing a documented risk-only fallback when its other required inputs are valid. An immediate cash floor is externally supplied and must be justified using the liability calculations; the bridge does not infer a ten-year funding guarantee from short-horizon forecast uncertainty.

7.2 Scenario mathematics

The scenario generator supports Gaussian log increments and a bounded heavy-tail alternative. Given an explicitly supplied annual drift vector a, annual covariance Σ, and horizon Δ, the Gaussian branch starts with

$$Z = (a - \tfrac{1}{2}\operatorname{diag}\Sigma)\Delta + \sqrt{\Delta}\, L\varepsilon, \quad LL^\top = \Sigma, \quad \varepsilon \sim \mathcal{N}(0, I). \tag{19}$$

The heavy-tail branch multiplies the common Gaussian shock by $\sqrt{(\nu-2)/U}$ for $U \sim \chi^2_\nu$, preserving finite increment covariance when $\nu > 2$. Optional shared Poisson-normal jumps represent declared systemic stress. Log increments are clipped to a recorded bound before conversion to simple returns. Even the Gaussian branch is changed by this cap; clipping frequency is therefore part of its metadata. These are hypothetical scenarios, not calibrated recession probabilities or exact samples from the neural marginal density.

Bond stress returns use an explicitly declared duration–convexity approximation, carry, and spread changes. Large yield moves, callable instruments, default, and changing cash-flow schedules can invalidate the local approximation. A real bond portfolio needs cash-flow-aware instruments and suitable issuer data. The mandate module includes liability cash-flow matching through linear programming when instrument cash flows are supplied.

7.3 Self-financing objective and restrictions

Let $w_i \geq 0$ denote risky weights as fractions of post-cost NAV, with cash $w_c = 1 - \sum_i w_i$. Let w^0 be current risky holdings relative to pre-trade NAV N, and let α be the fraction of NAV remaining after transaction costs. Traded fractions are $d_i = \alpha w_i - w_i^0$. With per-side rates c_i^+, c_i^-,

$$\alpha + \sum_i \{c_i^+ \max(d_i, 0) + c_i^- \max(-d_i, 0)\} = 1. \tag{20}$$

The implementation solves this equation before evaluating cash and traded-dollar limits. For scenario simple returns r_s and cash return r_c, the net outcome is

$$R_s(w) = \alpha\{1 + w^\top r_s + w_c r_c\} - 1. \tag{21}$$

The research objective combines predicted simple-return reward, a covariance penalty, turnover, and optional CVaR. CVaR can be represented using a threshold and upper-tail excess losses [17]; finite weighted scenarios require fractional probability mass at a tail boundary. The package computes that tail mass explicitly.

The feasible set enforces the mandate's eligible universe, allowed markets and asset classes, security weight caps, group weight floors and caps, cash bounds, foreign-currency exposure, turnover, and dollar participation limits based on known average daily traded value. An absolute immediate-cash reserve requires $\alpha N w_c \geq B$; that constraint alone does not certify future operating liabilities. Instrument-level foreign ownership rules and country portfolio caps are different quantities and are never interchanged. Fees, liquidity, costs, and the common forecast horizon must be explicitly supplied or declared as research assumptions.

The solver's success flag is insufficient. The proposed weights are independently checked against the same constraints. Fallbacks include keeping feasible holdings, a constrained risk-only allocation, or an eligible cash preference. Each candidate faces the same feasibility checks. If no fallback is feasible, the result is explicitly infeasible; it does not manufacture an apparently safe portfolio by violating a cash floor, mandatory sector floor, nontradable holding restriction, or turnover restriction. The instrument interface supports a dated tradable flag, not a complete settlement-accounting engine. Lot rounding requires another constraint check because continuous feasible weights need not survive rounding.

7.4 Recession and fallback policy

An anticipated recession is a scenario, not an established premise. The provided regime policy accepts a separately supplied recession probability, observed drawdown, volatility, data quality, forecast availability, and calibration status. Example development thresholds distinguish normal, defensive, cash-preference, and no-action states. They are configurable team policy, not legal requirements or estimated optimal boundaries. The package does not claim an autonomous, validated macroeconomic recession detector.

Data-quality or calibration failure leads to no action; unavailable forecasts can trigger a constrained risk-only fallback. A defensive setting raises risk aversion and may prefer a larger cash share while retaining hard mandate feasibility. Its value must be tested against a static policy. Transaction costs and missed recoveries can make defensive timing worse, so a fixed diversified eligible allocation remains a necessary baseline.

8 Laura Gao case: funding before speculation

The available source is the user's preparation document containing a copied case study [20]. It is stronger evidence than an earlier assistant summary, but it is not represented as an independently authenticated organizer PDF. The package preserves the document identifier, revision, case-study tab, relevant text, and access date.

For a specified constant net annual reserve return $r > -1$, the beginning-2033 reserve for the payments is an annuity due:

$$B(r) = 50{,}000 \sum_{k=0}^{9} (1 + r)^{-k}. \tag{22}$$

Case requirement	Implementation interpretation
Beginning 2027	Contribution of \$300,000
Beginning 2028	Additional contribution of \$150,000
Before 2033	No other contributions or withdrawals in the stated projection
Beginning 2033	Establish operating-payment reserve before first payment and any facility contribution
2033–2042	Ten nominal \$50,000 payments, each at the beginning of its year
Facility contribution	A credible range is required in 2031; flexibility and operating obligations constrain the eventual amount
Outside resources	Excluded from meeting these operating obligations
WInS simulation results	Excluded from the long-horizon financial projection

Table 2: Recovered case facts. Numerical risk tolerance, a confidence target, exact market allocations, and a Taiwan portfolio cap are not supplied by this case.

The first payment is undiscounted because it occurs at the reserve date. At $r = 0$, the algebraic reserve is \$500,000. This is an accounting identity under the stated return assumption, not a current yield quote or a guarantee of funding. Using an end-of-year annuity would understate the needed reserve.

If annual returns for 2027–2032 are R_y, projected wealth immediately before beginning-2033 uses is

$$V_{2033} = 300{,}000 \prod_{y=2027}^{2032} (1 + R_y) + 150{,}000 \prod_{y=2028}^{2032} (1 + R_y). \tag{23}$$

For an explicitly chosen flexibility amount F, a mechanical facility ceiling is $\max(0, V_{2033} - B - F)$. It becomes a responsible proposed contribution only after accounting for the distribution of V_{2033}, the reserve instrument cash flows, funding reliability, and the client's preferences. The 2031 update conditions on the actual then-current balance and the remaining two years of scenario returns; it does not confuse a 2026 forecast with an updated 2031 commitment.

The case does not specify a numerical funding-confidence level, flexibility reserve, approved security list, or every trading restriction. Consequently, the default Laura allocation profile remains incomplete for instrument recommendations until these inputs are documented. The cash-flow arithmetic is available independently. The official Wharton FAQ directs participants to competition materials for detailed rules and the eligible universe [21]; a generic list of popular tickers cannot substitute for them.

The Taiwan example also illustrates why provenance matters. The TWSE guide describes a 10% daily price fluctuation band for specified products, with exceptions [19]. That is not a universal 10% foreign-investor portfolio or trading ceiling. A team may choose a 10% Taiwan exposure cap as its own policy, but the configuration labels it accordingly and requires its source and effective dates.

9 Prospective experimental design

9.1 Data and protected boundaries

The first study should use a licensed, point-in-time panel with documented total-return treatment and a stable executable-observation definition. A fund or ETF panel can make distribution accounting more tractable, but it still needs historical membership and closures; present-day survivors alone are not a valid substitute. Individual equities require delistings and corporate actions. Bond or bond-fund experiments need price or total-return series rather than yields as

asset returns. The same underlying asset panel is supplied to all compared methods.

The example configurations choose 1 January 2025 UTC as the protected-tail start before any dataset has been inspected in this session. This is a prospective candidate boundary, not an assertion about the available dataset. A different chosen history requires a recorded development protocol amendment before protected outcomes are viewed. Development commands filter out the protected tail before constructing windows and reporting statistics. Windows whose labels cross the protected boundary are also removed from development.

Training, validation, and development-test partitions use unique decision times. No random row split mixes adjacent windows. Purging removes samples whose label availability reaches the next partition, and an explicit embargo adds a gap. The default is five decision-time bars for a five-interval target. Purging does not eliminate every dependence; it prevents a specific label-overlap leakage route. Cross-asset dependence is handled during inference, not by pretending instruments on one date are independent.

9.2 Models and ablations

The primary candidate is FREM with memory, compared to its memory-off counterpart. Secondary mechanism comparisons use FIFO, uniform eviction, no spectral features, no JEPA loss, no reservoir, no n-gram descriptor, no feature gates, no adaptive controller, and fixed predictive scale. Classical and neural controls include zero expected log return, ridge, reservoir ridge, XGBoost, GRU, DLinear, and a compact patch Transformer.

Each architecture receives the same allowed context and outcomes for its declared comparison. Capacity-changing ablations report actual parameter counts and wall time. The memory-off model retains unused gate parameters, avoiding an artificial parameter-count benefit, while memory retrieval still has additional compute and storage. Equal parameter counts do not prove equal capacity. Baseline predictive scales use an inner chronological training calibration tail; refitting the point predictor afterward incurs extra work that is recorded. A strong efficiency claim requires explicit resource-matched outcomes, not only a shared nominal epoch count.

Low-data experiments take chronological suffixes containing 10%, 25%, 50%, and 100% of training dates while keeping evaluation endpoints fixed. This asks how the methods respond to fewer allowed training observations. It cannot establish reliable prediction from "almost no data" across arbitrary markets. Sensitivity axes include context length, reservoir size, spectral bands, memory capacity, horizon, and retention half-life. A fixed development budget bounds this search.

9.3 Robustness and distribution shift

Input-noise and missingness assays modify evaluation features with training-only scales and preserve targets. Missingness channels are updated where available. These assays test representation robustness, not realism of every market-data outage. OOD studies hold out declared markets, sectors, or chronological regimes. Held-out asset names must be fixed before the corresponding outcomes are inspected, and unknown assets must not accidentally reuse existing integer IDs.

Historical crisis and recovery subgroups require dates fixed independently of observed model performance. Real-time macro vintages are needed for any decision rule that would have used macroeconomic information at that time. FRED/ALFRED acquisition and availability conventions are documented separately [8]; date-only vintages receive a conservative delay unless actual release timestamps are available. Revised final-series recession labels are not causal input features.

Portfolio sensitivity includes spread, slippage, transaction fees, withholding and tax assumptions when applicable, FX, liquidity, participation, and cost multipliers. Predictive rows with overlapping horizons are not compounded as independent trades. The provided execution-ledger helper accepts synchronized nonoverlapping round trips and rejects asynchronous multi-market intervals. A more general multi-currency holdings simulator is a separate extension, not an invisible assumption of this release.

10 Statistics, evidence, and stopping rules

The primary endpoint is date-balanced Student-t negative log likelihood, a proper distributional score under its usual integrability conditions [11]. Each observation is scored using its saved, fitted location, scale, and degrees of freedom. First average complete predeclared optimization seeds per observation, repeated observations within each UTC date and instrument, instruments equally within a date, and dates equally. A seed measures optimization variability; it does not add an independent economic history.

Let $d_t = \bar{\ell}_t^{\text{control}} - \bar{\ell}_t^{\text{FREM}}$. Positive values favor FREM. The implementation uses a paired circular moving-date-block percentile interval and a centered-bootstrap null test. Consecutive date blocks preserve contemporaneous observations together. Block resampling is motivated by dependent observations [14]; changing regimes can still violate its assumptions. The candidate default is 20 dates, with sensitivity at 5 and 60 subject to the observed execution-and-maturity overlap lower bound, and 5,000 resamples with random seed 1901 for confirmation. A 95% interval entirely above 0.01 nats per observation is the prespecified practical-support criterion. This threshold is a study design choice rather than a universal scientific standard.

Exact matching of decision time, instrument, seed, realized outcome, label-availability time, and any supplied entry and exit times is required before pairing. Missing methods or seeds cause an incomplete-evidence result; inner-joining away failures is forbidden. Secondary mechanism hypotheses use the complete declared multiplicity family and Holm adjustment [12]. Their adjusted directional significance remains separate from the primary practical-effect criterion. Time-series dependence, limited crisis episodes, unmodeled changing opportunity sets, and few effectively independent blocks can still make inference inconclusive.

Secondary endpoints include proper distribution scores, direction probability and Brier score, interval coverage and width, MAE, date-level rank association, net wealth, drawdown, turnover, funding shortfall, actual runtime, parameter counts, memory occupancy, and resource measurements. Seed standard deviation in a descriptive figure is not a confidence interval over market histories. Sharpe and annualized quantities require a justified cadence and nonoverlapping returns; they are not produced by multiplying arbitrary horizon scores by 252.

Before confirmation, freeze code and source hashes, raw data and provenance-manifest hashes, feature schema, protected boundary, model variants, hyperparameters, seed family, primary endpoint, practical threshold, exclusions, stopping rules, and environment. Verification rejects changed identities. Each run archives the exact analysis protocol, and confirmatory analyses verify its frozen hash and settings. A consumption receipt prevents accidental reuse of the same confirmatory freeze, including after interruption. This is an operational research guard, not protection against an owner deliberately altering their files. Protected outcomes cannot be used to revise that version's central hypothesis. A new scientific change becomes a successor version with the original result retained.

The valid terminal categories are supported, mixed, negative, and inconclusive. Invalid is an evidence-quality state requiring an explicit defect description and corrected protocol. A valid negative result is accepted; repeated evaluation solely to obtain a favorable seed is prohibited.

The current state remains **not evaluated**, which is different from an empirical inconclusive result.

11 Computational complexity and engineering limits

For one context, a dense reservoir costs $O(L(R^2+RD))$, the spectral transform $O(DL\log L)$, and a GRU approximately $O(L(d^2+dD))$. Spectral pooling and fusion add costs proportional to the actual descriptor dimensions. Exact memory retrieval costs $O(CD_k + C\log C)$ with the current full candidate ordering, plus $O(K)$ summary work. Pending-queue scans are $O(P)$ per advance in the straightforward implementation. These terms can dominate before neural compute on small models.

No claim of improved tokens per second, training speed, or energy use is made. A dense fixed recurrence reduces trainable parameters in one branch but still performs recurrent matrix multiplication. A FIFO comparison and a no-memory comparison help reveal whether predictive gains justify lookup overhead. Optimizations such as approximate search, heap-based maturity queues, and batched fixed-key caching may be useful successors, but they require equivalence or changed-method evaluation before replacing the frozen implementation.

The default device is CPU, appropriate for a first bounded compatibility run on a modest machine. Runtime has been measured on bounded synthetic and public exploratory exercises; these timings do not establish comparative efficiency or market-scale feasibility. PyTorch's complex FFT path is not assumed portable to every accelerator backend; the trainer explicitly rejects unsupported spectral MPS use rather than silently changing its computation. Checkpoints preserve model buffers, EMA targets, optimizer, scaler, memory, controller, schemas, and fitting cutoffs. Baseline persistence follows the relevant estimator format, and pickle-derived files must be treated as trusted local artifacts only.

Experiment directories are unique, retain partial outputs, record failures and interruptions, and include environment and source identities. Process peak RSS is explicitly a process-lifetime high-water mark rather than isolated per-model memory. Cooperative wall-time limits are checked during training and between matrix cells. A single native tree fit is not forcibly interrupted by that API; a scheduler-level hard timeout is needed when hard resource containment matters.

12 Adversarial audit and required evidence

Objection	Mechanism or cause	Repair and required evidence
The combination is not novel	All major ingredient families predate this study	Restrict novelty to the retention hypothesis; compare direct neighbors and do not claim a new Fourier transform
Memory stores noise	Rare events need not recur or predict returns	Compare utility, FIFO, uniform, and no memory; inspect delayed utility and retrievals
Memory exploits hindsight	Future labels or revised data enter too early	Timestamp gates, vintage provenance, same-time grouping, maturity tests, protected tail
JEPA collapses or hurts	Predictable latent features may become trivial or irrelevant	Latent variance/covariance diagnostics, anti-collapse losses, JEPA-off control

Objection	Mechanism or cause	Repair and required evidence
Low-data claims are overstated	Priors cannot recover absent signal	Fixed low-data curves, zero/ridge controls, uncertainty, accepted null outcomes
Spectral artifacts dominate	Windowing, irregular sampling, changing price scale	Return-based features, trailing DFT, explicit observation-time meaning, window/horizon sensitivity
Residual memory becomes stale	Neural predictor and market drift	Fixed keys, age decay, utility credit, residual-drift analysis; no guarantee of stationarity
Learning controller just tunes more	Extra validation reuse favors one method	Fixed policy and budget, controller-off comparison, report all tuning information
Trees explain the win	Additional model capacity or calibration labels	Standalone XGBoost, separate hybrid comparison, matched-information interpretation
Uncertainty is misused	Scale confused with SD or log mean with price mean	Save degrees of freedom, exact scoring, calibration, explicit bounded simple-return scenarios
Funding reserve is optimistic	Constant-rate assumptions hide sequence risk	Annuity-due timing, scenario reserve paths, cash-flow matching, no outside-funding assumption
Cross-market backtest is unrealizable	Different close times, holidays, FX, and fills	UTC availability, synchronized ledger restriction, explicit portfolio-engine extension
Defaults disappear	Log returns cannot represent zero recovery	Fail and record the defect; add a default-mass model in a successor rather than exclude failed issuers
Good results are selection artifacts	Many variants and repeated holdouts	Frozen protocol, complete seed family, date blocks, multiplicity and failure retention
Fallback violates rules	Cash or reduced turnover is assumed always feasible	Recheck all hard constraints for every fallback and after lot rounding

13 FREM-S14 prospective successor

FREM-S14 is a separately versioned development successor. It does not reopen or replace the completed S2–S8 exploratory studies, and no S14 result is pooled with their evidence. The central prospective question is whether a learned causal router over multiscale reservoir, spectral, stochastic-latent, and outcome-matured engram features can improve a proper held-out distributional score over matched mechanism-off and simpler controls. The implementation is complete enough to state and execute the declared engineering and historical proxy protocol, but the protocol has not produced a qualified market result.

13.1 Forward mechanism

For each leak $\alpha_j \in \{0.1, 0.4, 0.9\}$, the active reservoir coordinates follow

$$r_t^{(j)} = (1-\alpha_j) r_{t-1}^{(j)} + \alpha_j \tanh(W r_{t-1}^{(j)} + U x_t + b), \qquad \|W\|_2 \le 0.85. \tag{24}$$

Inactive coordinates are masked. Capacity is fixed; at declared training epochs, inactive units may be recruited by stable ranking of their absolute sample correlation with training residuals. Counts, sums, squares, and cross-products are checkpointed. Validation and evaluation labels cannot change the active set. The `no_neurogenesis` and `fixed_full_capacity` arms distinguish recruitment from mere capacity.

The latent vector is arranged on a periodic 4×8 lattice. With positive row and column diffusion coefficients, damping $\kappa > 0$, lattice Laplacians L_r, L_c, and $A = D_r L_r + D_c L_c + \kappa I$, S14 uses the conditional linear Gaussian transition

$$dZ = (-AZ + f(z_0))\, dt + \sigma\, dW, \qquad f(z_0) = \tanh(Bz_0 + c). \tag{25}$$

In the orthonormal eigenbasis, a mode with eigenvalue $a > 0$ has

$$\mathbb{E}[Z_{t+\Delta} \mid Z_t = z] = e^{-a\Delta} z + \frac{1 - e^{-a\Delta}}{a} f, \quad \mathrm{Var}[Z_{t+\Delta} \mid Z_t] = \sigma^2 \frac{1 - e^{-2a\Delta}}{2a}. \tag{26}$$

The implementation uses `expm1` forms near zero. This is exact for the declared finite-dimensional conditional model; it is not a physical market PDE or a convergence theorem for a nonlinear continuum process. The `no_spde` arm replaces it with the registered MLP transition.

Masked online contexts are predicted at the distinct offsets $\lfloor H/2 \rfloor$ and H. EMA stop-gradient targets consume shifted trailing contexts, and the maximum offset governs purging. Forecast loss is combined with mode-wise Gaussian latent likelihood, cosine agreement, variance floors, and covariance penalties. The `no_mask`, `no_multihorizon`, and `no_jepa` arms isolate these choices.

Memory keys combine fixed statistical, Fourier, and short n-gram coordinates. An eligible episode must have the same asset identity and outcome maturity strictly before the current decision. A bounded top-k search uses cosine and age weights, followed by one convex personalized graph-diffusion step among the eligible selected neighbors. Invalid pairs are zeroed before moments and the weights are renormalized. The full-arm trace candidate retains episode decision and maturity times, asset, slot, similarity, age, value, base weight, graph weight, final weight, coverage, and reconstructed memory moments. This trace is an audit mechanism, not evidence that retrieval helps. The `no_memory`, `no_graph`, and `no_ngram` controls remove the corresponding paths.

Three capped-simplex gates separately weight experts for likelihood, downside Brier loss, and tail pinball loss. A balance penalty discourages collapse. The forecast law is a finite-variance Student mixture whose variance includes both within-component variance and centered between-component dispersion. The law can assign mass below a -100% simple return and is therefore a scoring model, not a solvency-safe return simulator. Portfolio replay uses separately bounded finite scenarios and never treats the unbounded predictive law as executable wealth support.

13.2 Training, controls, and fixed proxy matrix

Training uses AdamW, gradient clipping, deterministic registered seeds, cosine scheduling, early stopping on the validation partition, CUDA checks, and exact epoch resume. The fixed arm family is

> full, no-memory, no-JEPA, no-neurogenesis, no-reservoir, no-spectral, uniform-router, no-risk, zero, ridge, XGBoost, hybrid-XGBoost, fixed-full-capacity, no-ngram, GRU, no-SPDE, no-multihorizon, no-graph, and no-mask.

The registered seeds are 11, 29, and 47. There is no automatic hyperparameter search in this matrix. Hybrid XGBoost is validation selected and its additional compute must be reported rather than hidden.

The historical proxy cohort is SPY, QQQ, IWM, DIA, TLT, IEF, GLD, and SLV. Three expanding prefixes end before 2011, 2015, and 2018. The intake requires identical calendars and rejects intersection selection, forward filling, and asset substitution. Nine causal historical features create 20-observation compounded targets. Each fold uses chronological 60% training, 20% validation, and 20% evaluation, with a 20-observation embargo and outcome-maturity purge. The complete matrix therefore has $19 \times 3 \times 3 = 171$ declared cells. Those cells are not 171 independent financial histories. A fold may consume at most 2,800 seconds and the notebook at most 13,500 seconds including tests and qualification. A failed, missing, or timed-out cell stays in the denominator.

The primary endpoint is the complete-family date-balanced Student-mixture negative log likelihood. Seeds are averaged within arm and fold before equal fold aggregation. Paired circular 20-date blocks are resampled 2,000 times; the complete arm family receives Holm adjustment. The practical improvement margin is 0.01 nats per observation. Independent reconstruction checks raw laws, moments, scores, allocation targets, self-financing wealth, fees, drift, and liquidation. Even a completed favorable proxy outcome would remain exploratory because the source is a present-day adjusted-close archive without verified point-in-time adjustment vintages, inactive securities, executable quotes, spreads, impact, or terminal distributions.

13.3 Implementation and execution evidence through 20 October 2026

The current trace candidate is package 0.1.5. Its source identity is `6ccbee61b0518d0e fa297bcfe547e769c4e3a9c51610341f4e8c622258414d37`. The immutable release ZIP, wheel, and notebook hashes are retained in the repository release receipt. In the current audit, 57 dependency-free tests passed from source and the same 57 passed from a fresh wheel installation under Python 3.12. Scoped Ruff formatting and lint, bytecode compilation, an offline lock check, and exact release-hash readback passed. These checks exclude 46 declared numerical contracts and all CUDA/AMP execution.

Two earlier private GPU attempts are scientifically non-results. Version 1 ran 41 tests and failed one strict float32 reservoir-contraction check before qualification or market access. Its code, output, and failure receipt are preserved. A prospective normalization repair retained the contraction bound and acceptance threshold. Saved version 2 used the repaired 0.1.3 release but Kaggle exhausted 30 dataset-mount retries before notebook code ran; its visible log reports zero output bytes. Consequently it executed zero tests, zero qualification arms, and zero historical cells. This external infrastructure failure neither supports nor falsifies the model and permits an exact, recorded retry without outcome-driven modification.

The admissible continuation is narrow: preserve both failed saved versions; verify the exact 0.1.5 notebook and release hashes; obtain a GPU runtime with the declared dataset mounted; complete all numerical tests and generated qualification; then execute or explicitly fail every one of the 171 cells. Only a downloaded, independently verified complete bundle can support proxy analysis. A valid negative result closes this version. A genuine validity defect requires a recorded corrected successor; a favorable seed, fold, block length, or subgroup cannot be chosen after outcomes are seen.

Artifact or stage	Status	Evidence boundary
S14 0.1.5 source and wheel	57 + 57 tests pass	Dependency-free contracts only; not numerical or CUDA qualification
GPU saved version 1	Engineering failure	41 pass, one contraction failure; zero market cells
GPU saved version 2	External mount failure	Notebook never ran; zero-byte output and zero market cells
Generated 19-arm qualification	Not executed for 0.1.5	Required before data intake
Historical proxy matrix	0/171 complete	No S14 predictive or economic result
Protected 2025 onward boundary	Not accessed	No confirmatory claim or consumption

Table 4: FREM-S14 implementation and execution ledger. Engineering evidence is not promoted to model efficacy.

14 Current evidence and failure analysis

14.1 Engineering execution and exact scope

The original no-execution authoring snapshot is preserved separately. The later retained runtime receipt reports 163 FREM tests, 67 RE-JEPA v0.3 tests, and 45 RENG-JEPA v0.1 tests passing with deprecation warnings treated as errors. These counts are historical receipts, not newly rerun tests in this manuscript audit. Ten subsequent synthetic integration cells completed: four FREM/control cells, two RE-JEPA cells, and four RENG-JEPA regimes. The successor architectures are separate studies, so their diagnostics are not pooled with FREM or counted as independent replications.

The FREM fixture comprises four generated Gaussian total-return series calibrated using the full retrospective Yahoo source sample. Its availability times are generated, its assets are synthetic, and its calibration is not point-in-time. Seed 11 and two neural epochs were selected for integration checks. The retained train/validation/test split has 1,156/364/368 rows, with 92 evaluation dates. This is not the full development or confirmatory matrix. The fixture-specific 2020 cutoff is distinct from the still-unaccessed canonical market boundary of 1 January 2025.

Model	NLL	MAE	Direction accuracy	Fit seconds
frem	-0.9014	0.0539	0.4701	6.23
no_memory	-1.4663	0.0418	0.5408	7.13
ridge	-1.6915	0.0335	0.4457	2.71
xgboost	-1.7904	0.0307	0.4946	0.74

Table 5: Retained synthetic integration run only: four generated assets, 92 dates, 368 paired predictions per model, seed 11, two neural epochs. Lower NLL and MAE are better. Timings are the original single-run measurements, not efficiency evidence.

The recomputed control-minus-FREM NLL difference is -0.564927 nats. The retained 95% interval is [-1.132572, -0.002588], with centered-bootstrap two-sided $p = 0.054726$. The original analysis used 200 resamples and 20-date blocks; it is unchanged by this audit.

14.2 Interpretation of the negative diagnostic

FREM has worse NLL and MAE than its memory-off control on this fixture, and both classical controls have lower NLL. The result does not support an improvement claim even in this limited

setting. The negative value is preserved; no thresholds, bootstrap blocks, seeds, or fit settings have been changed in response. The percentile interval and centered-bootstrap test are different procedures with only 200 resamples: the interval's slightly negative upper endpoint does not justify describing the reported two-sided test as significant at five percent. Neither procedure supplies a market conclusion.

Plausible explanations include noise retrieval, short-run optimization, and distribution miscalibration; this exercise does not identify which explanation is responsible. The generator and short training schedule were chosen for engineering compatibility, not as a controlled test of economic recurrence. A discriminating development study must compare matured-utility retention with no-memory, FIFO, and uniform retention under identical admitted observations and budgets. That study requires an admissible dataset; it has not been executed, and the present outcome does not authorize retuning a protected result.

14.3 Independent calculation and traceability

The standard-library audit in `scripts/audit_engineering_evidence.py` recomputes Student-t NLL from each saved location, scale, degree of freedom, and outcome. It reproduces equal-asset within-date and equal-date aggregation, verifies the complete configured family and exact paired outcomes, checks execution/maturity ordering and split metadata, and verifies source, dataset, prediction, and receipt hashes. It imports no model package and performs no fit. The unchanged bootstrap interval is copied from the hash-verified original analysis; it is not presented as independently recomputed. A separate set of eight small audit tests checks an analytic Cauchy case, scale normalization, uneven observation weighting, and rejection of malformed, duplicate, or out-of-bound evidence.

The evidence chain is retained as `research/evidence_audit_20261011_v2/audit.json`; its generated table is included verbatim here. The canonical integration receipt is `research/all_admissible_runs_2026-10-11.json`. All original predictions, fitting budgets, configurations, protocol snapshots, partial failures, and successor receipts remain available. This numerical audit is not clean-environment model reproduction or an independent scientific review.

14.4 Public historical crisis campaign

A separate exploratory adapter has acquired histories for 24 present-day listings across 12 markets, ending before 2025. It declares six historical windows and four models, with at most 24 cells and a twelve-hour total ceiling. Current-vintage adjustments, survivor selection, assumed monthly timing, and absence of a common FX/NAV convention restrict it to exploratory local-currency price-proxy analysis. It is not the canonical point-in-time panel and cannot establish a complete global-equity or investment result.

The first admission failure remains preserved. Later authorized slow-mode execution completed the public matrix; its actual results and the separately frozen S2/S3 studies follow.

15 Completed public exploratory studies

These studies are distinct from the proposed canonical confirmation and from the synthetic fixture above. They use public current-survivor histories, current-vintage adjustments and research-coordinate timestamps. They do not establish historical eligibility, executable global holdings, or investment profitability. No protected 2025 onward outcomes were accessed. Valid completed protocols are closed; no favorable arm replaces a predeclared primary.

15.1 S2: improved fitting without a memory advantage

The S2 campaign completed 22 learned fits and 24 prediction cells on 23 survivor stocks, 19 crisis months and two optimization seeds. Lower Student-t NLL is better. These are date-balanced descriptive scores; no significance test was performed. Both seeds selected zero residual-memory correction.

Model	NLL
legacy_frem	-0.490295
legacy_no_memory	-0.481158
standardized_frem	-0.559094
standardized_no_memory	-0.558770
frozen_residual	-0.603182
frozen_no_memory	-0.603182
zero	-0.709526
ridge	-0.425342
xgboost	-0.453538
gru	-0.325352
dlinear	-0.383834
patch_transformer	-0.392630

Scaling improved optimization relative to legacy FREM, but the zero-return control remained best. The frozen residual and memory-off successor are identical in score. This does not support a memory-benefit or state-of-the-art claim.

15.2 Six-window public slow study

All 24 declared fits completed: six windows, four methods, seed 11 and five neural epochs. An explicitly recorded whole-month missing-data amendment was used; the strict original zero-quote rejection and earlier failures remain preserved. These single-seed comparisons have exploratory unadjusted uncertainty. Positive gain below is no-memory NLL minus FREM NLL.

Window	FREM	No memory	Gain
great_recession	-0.489055	-0.451771	+0.037284
recovery	-1.196772	-1.161821	+0.034950
year_2011	-1.235244	-1.231032	+0.004212
years_2015_2016	-1.115238	-1.082327	+0.032911
year_2020	-0.784608	-0.792614	-0.008006
year_2022	-0.981086	-1.019331	-0.038245

FREM improved point NLL in four windows and worsened in 2020 and 2022. There is no pooled confirmatory or profitability conclusion.

15.3 S3: primary memory result and uncertainty

S3 freezes a calibrated base and applies causal delayed, versioned residual replay. It completed 24 learned public fits, 12 constant-baseline cells and 15 generated mechanism cells. The public sample contains twelve US current-survivor stocks, with 144 asset-month outcomes in January–December 2023 and 120 in January–October 2024. Seeds 11, 29 and 47 are optimization repeats, averaged before date-level inference. The primary remains combined memory versus the identical calibrated memory-off base. The practical threshold is 0.01 nats.

Window	Gain	Exploratory 95% interval	Months
2023	-0.014116	[-0.036455, +0.006302]	12
2024	+0.007006	[-0.007940, +0.022561]	10

Both three-month block intervals include zero and fail the practical-support criterion. The declared family retains Holm adjustment. The 2024 six-month sensitivity is unavailable because ten months cannot supply two six-month blocks; no block was shortened to force a result.

15.4 Complete S3 memory arms

The following gains use the same calibrated checkpoint with memory off as control. No tested arm improved both windows. Secondary point gains do not replace the primary comparison.

Arm	2023 gain	2024 gain
off	+0.000000	+0.000000
frozen	+0.000000	+0.000000
recent	-0.000705	+0.000000
random	-0.010605	+0.006841
mean	-0.010366	+0.008279
risk	+0.002436	-0.001244
combined	-0.014116	+0.007006
no_confidence	-0.021568	+0.003633
balanced	-0.014116	+0.007006
larger	-0.005289	+0.012700
dual	-0.006473	+0.013683
cross_asset	+0.007388	-0.001506
learned	-0.006754	+0.000564
utility	-0.002627	+0.002309

15.5 Base controls and calibration

Model	2023 NLL	2024 NLL
frem/off	-1.345176	-1.437064
no_jepa/off	-1.349918	-1.447446
ridge/off	-1.255116	-1.247335
xgboost/off	-1.209832	-1.334934
zero/off	-1.254592	-1.361529
historical_mean/off	-1.271172	-1.393529

Raw FREM before extra calibration scored -1.354994 and -1.455920. Extra calibration worsened both windows. The no-JEPA control also had better point scores than calibrated FREM in both windows. These observations do not establish causal attribution or authorize retrospective primary selection.

15.6 Mechanism evidence and failure diagnosis

Generated recurrence demonstrates behavior under planted information, not financial predictability. Its retrieval keys directly include the planted regime state. The null showed

small harm within the declared tolerance. In real S3 predictions, retrieval and nonzero gates execute; selected corrections fail to transfer consistently. A later descriptive audit identifies tiny validation advantages, old base/calibration blocks and nearly flat within-neighbor similarity weighting. For unit-normalized 103-dimensional keys, mean squared distance is at most 4/103, so $\exp(-d)$ is at least 0.9619. This is a scale-design concern, not proof that the completed experiment is invalid. A successor must test altered distance scaling and recent chronological fitting separately, retaining zero correction and matched-random controls.

15.7 Provenance and evidence boundary

Tables are generated directly from retained S2, slow-study and S3 JSON summaries by `research/checklist_completion_20261012/generate_paper_results.py`. Their input identities are saved in `research/checklist_completion_20261012/table_sources.json`. Raw predictions, freezes, calibration choices, failed attempts and source archives remain in their original run directories. The S3 receipt records 217 regression passes, not a fresh test run for this paper edit. Independent score reconstruction is distinct from retraining. Canonical confirmation, independent training reproduction, broad robustness and executable portfolio evidence remain incomplete.

15.8 S4: conservative development successor

After S3 was closed, a separately declared S4 development study reused the already inspected 2023/2024 public windows. It is not a fresh generalization test. Six recent-base fits (three seeds per window) generated 24 prediction cells. The base used the most recent 96 pre-validation dates, divided into 48/12/12/24 fitting, checkpoint-selection, calibration and residual dates with strict maturity purges, and five training epochs. All arms shared this base and mean-only correction. The distance intervention multiplies keys by $\sqrt{d}$, exactly converting the inherited mean squared distance to squared Euclidean distance. Temperature is fixed at one.

The four arms are memory off, original distance, Euclidean distance and matched random. Nonzero weights from $\{0.25, 0.5, 1\}$ were eligible only if each chronological half of validation improved NLL over zero correction by more than 0.01 nat. Every non-off arm in all six seed/window cells selected zero. Consequently all four arms scored the same NLL: -1.332779 in 2023 and -1.411278 in 2024. No memory improvement was demonstrated. Because the gate is zero, deployed equality does not establish that distance scaling itself is ineffective.

All 247 pre-fit tests passed. The campaign took 64.76 seconds and reached 401.84 MiB sampled worker RSS under the explicitly renewed single-worker slow-mode exception. An independent arithmetic audit reconstructed 3,168 rows to 2.0×10^{-15} and verified 81 frozen source/data identities. These are development and engineering observations, not current market forecasts, protected confirmation or independent retraining. Source: `outputs/frem_s4_20261012_attempt1/completion.json`; audit: `research/frem_s4_20261012/audit.json`. Completed S2/S3 results and all predecessor artifacts remain preserved.

15.9 Additional public regional benchmarks

Following authorization to collect publicly accessible data, seven official Kenneth French Data Library monthly factor files were downloaded: the US reference, developed, developed excluding US, Europe, Japan, Asia Pacific excluding Japan, and North America. The regional series begin in July 1990 and the US series in July 1926. The normalized table contains 3,666 region-month rows through December 2024, including 2008 for each series. Numeric fields on later-dated

rows are excluded before parsing. Retrieval timestamps, provider last-modified headers, original archive hashes, and normalized-file hashes are retained.

The regional documentation defines simple US-dollar returns including dividends and capital gains, with market excess returns relative to the US one-month Treasury-bill rate [10]. Percent units are divided by 100; the market return is reconstructed as market excess plus the reported risk-free return. These are current-vintage research portfolios, not individual-stock price histories or certified executable holdings. Regions overlap, publication vintages are not established, and no model was fitted to these new data. Acquisition broadens future exploratory coverage without filling the canonical data-admission gate. Raw and derived downloads remain local pending redistribution-rights verification.

15.10 Unresolved research and release gates

The canonical market hypothesis remains `NOT_EVALUATED`. Bounded public development and S3 component ablations are complete within their exploratory scope; the full canonical robustness, OOD, scaling and efficiency program remains incomplete. No confirmatory freeze or protected use has occurred. The original worldwide-equity request also requires historical inactive securities and terminal outcomes, a default-aware likelihood, and a validated global holdings/FX accounting convention. No valid negative market conclusion can be inferred from absent evidence.

Submission readiness additionally requires a completed evidence-based scientific disposition, full equation/code and literature review, clean-environment reproduction, author and affiliation confirmation, source/case redistribution rights, and final venue processing. Local TeX compilation and source packaging address document mechanics only. The original reported negative APEN/graph-memory evidence, invalid legacy experiments, and spent lockbox are preserved and are not rerun by this revision.

16 Conclusion

The completed public exploratory evidence does not establish the primary memory advantage. Extra calibration and memory selection did not transfer consistently. A negative or inconclusive result is a valid research outcome; no stock recommendation follows from these NLL tables.

FREM defines an implementable financial forecasting hypothesis: stable trailing descriptors can retrieve a bounded set of episodes whose retention responds to matured predictive utility. The model incorporates temporal latent prediction, fixed recurrent features, spectral weighting, motif keys, restricted adaptive learning, and explicit uncertainty. The allocation system treats legal eligibility, costs, and liability funding as separate constraints on how forecasts may be used. The mathematical guarantees concern causal access, bounded memory, and reservoir forgetting under stated assumptions; they do not prove returns, low-data success, or efficiency. The retained synthetic diagnostic favors memory-off, while the intended market question remains unanswered. The next scientific decision depends on admitted data, resource readiness, and the declared development and freeze process; the present draft makes no profitability, efficiency, or novelty-success claim.

A Implementation map

Module	Responsibility
`data.py`	Schema, provenance validation, causal features, windows, unlabeled inference, train-only scaling
`data_sources.py`	Explicit macro-vintage acquisition and provenance-preserving loading
`splits.py`	Purged chronological and walk-forward partitions
`model.py`	Temporal encoder, reservoir, spectral operator, return density, JEPA loss
`memory.py`	Timestamped episode retrieval, pending feedback, utility, eviction
`controller.py`	Declared learning schedules and matured expert weighting
`training.py`	Chronological fitting, EMA, validation, inference, checkpoints, residual tree
`baselines.py`	Classical, tree, recurrent, linear-decomposition, patch sequence controls
`metrics.py`, `evaluation.py`	Distributional scores and realizable nonoverlapping return accounting
`statistics.py`	Complete-seed aggregation, date-block inference, multiplicity
`mandate.py`	Rules, provenance, eligibility, cash flows, reserves, facility analysis
`risk.py`, `portfolio.py`	Explicit scenarios, covariance, regime policy, constrained allocations and fallbacks
`decision.py`	Dated forecast/horizon contracts and bounded scenario conversion into one research allocation
`experiments.py`, `robustness.py`	Bounded matrices, ablations, data-size and perturbation studies
`governance.py`, `artifacts.py`	Freezes, identities, unique runs, failure records, decision journal
`reporting.py`, `cli.py`	Evidence-derived outputs and explicit user commands

B Research-stage accounting

The idea and mathematical implementation contract are specified. Source code for the model and methodology is supplied. Runtime compatibility, bounded synthetic integration, and the S2, six-window and S3 public exploratory campaigns have receipts. Admitted-market dry-run readiness, scientific development, complete ablations, robustness, OOD, scaling, confirmatory freeze, protected evaluation, scientific evidence locking, independent reproduction, and final empirical release remain incomplete. Their future outputs are not implied by this manuscript's existence. The canonical state must be updated after each actual execution, retaining failures and superseded hypotheses.

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